3. GATE-FIRST STRUCTURE AND PROCESS EDITOR IMAGES

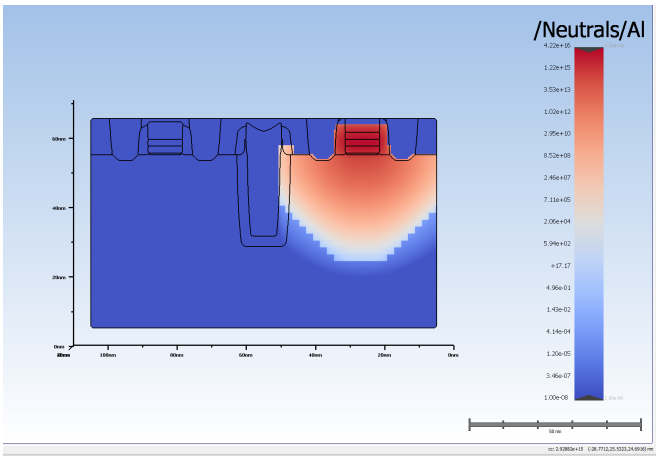


Figure 1 : Al Metal Diffusion in Gate-First Structure

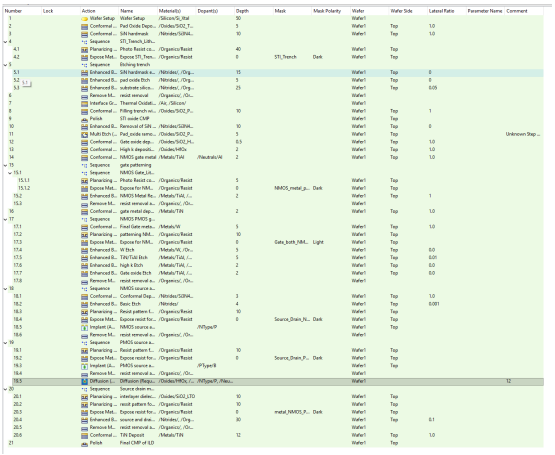


Figure 2 : Process Editor Steps for Gate-First Flow

3. ANNEAL TIME VS. AL CONTAMINATION PROFILE

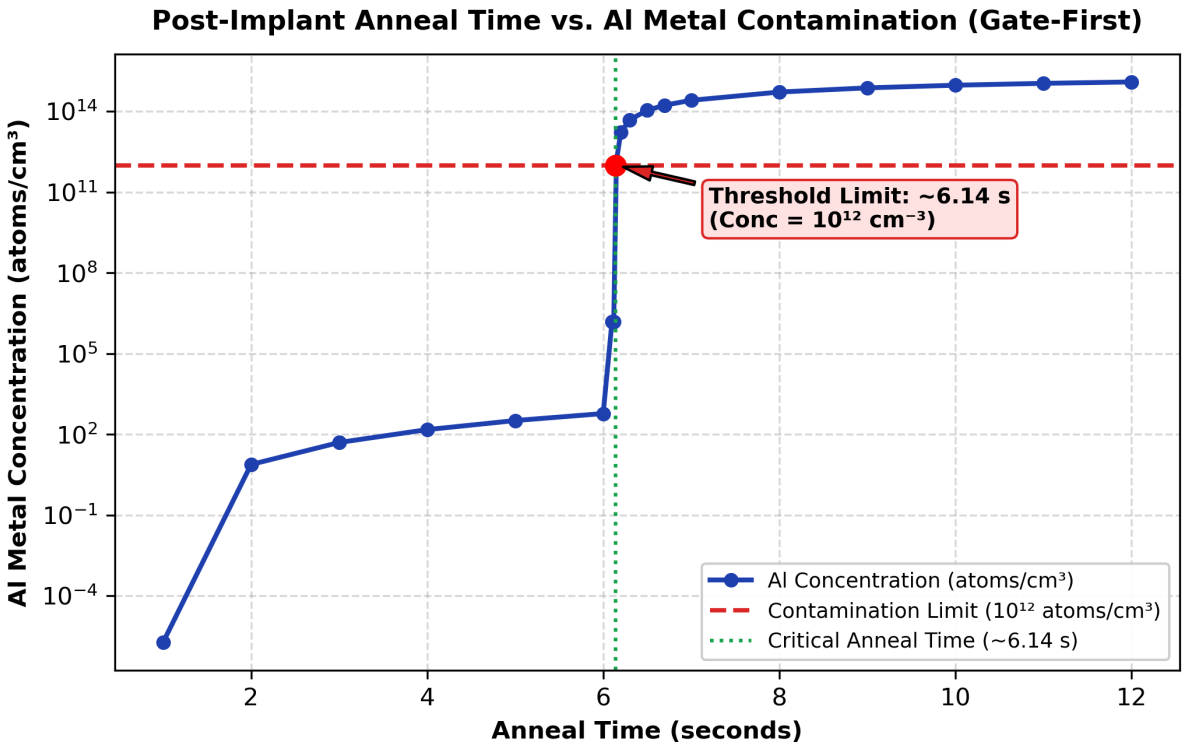


Figure 3 : Post-Implant Anneal Time vs. Al Metal Contamination

3. GATE OXIDE THICKNESS VS. AL CONTAMINATION PROFILE

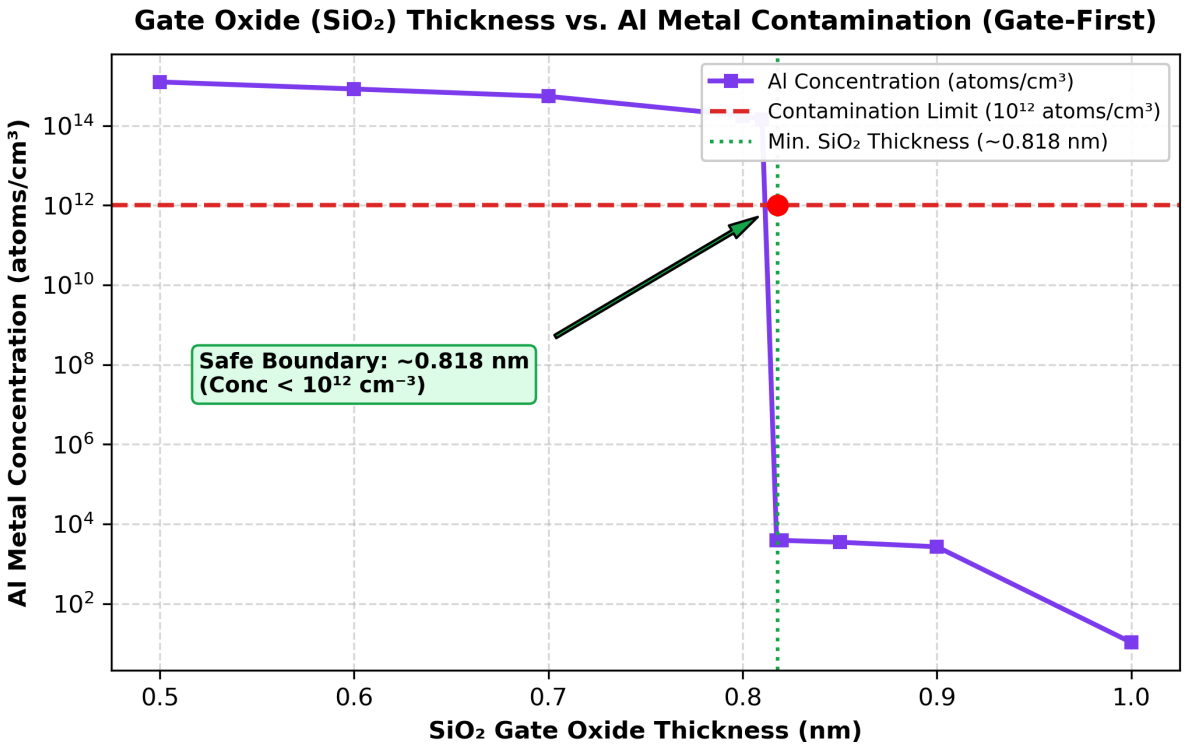


Figure 4 : Gate Oxide Thickness vs. Al Metal Contamination

4. GATE-LAST STRUCTURE AND PROCESS EDITOR IMAGES

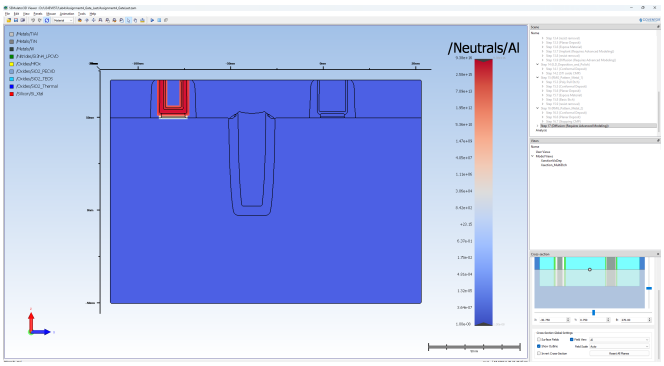


Figure 5 : Al Metal Diffusion in Gate-Last Structure

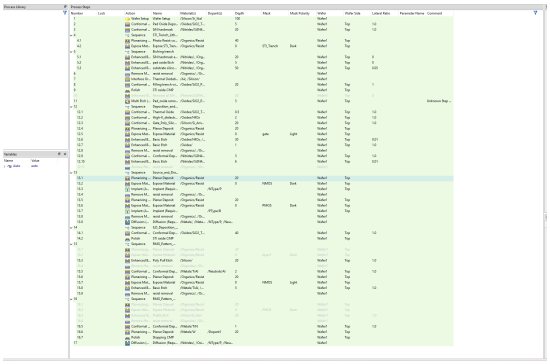


Figure 6 : Process Editor Steps for Gate-Last Flow

5. QUESTION 3: INBUILT 14 NM FINFET PROCESS ANALYSIS (14NMFINFET.ZAM)

The industrial 14 nm FinFET process project (14nmFinFET.zam) was examined in Semulator3D to analyze gate stack integration in modern 3D multi-gate transistors.

(a) Identification of Gate Deposition Steps:

In the 14nmFinFET.zam process flow, the gate module is implemented as follows:

• Step 2.5.1 (DEP SacOx): Conformal deposition of sacrificial gate oxide over the patterned 3D silicon fins.

• Step 2.5.3 (DEP SacGate): Deposition of sacrificial polysilicon dummy gate filling between the fins (Figure 7).

• Step 2.5.4 (CMP SacGate): Planarization of the sacrificial dummy polysilicon gate.

• Step 2.5.10–2.5.19: Lithographic patterning and anisotropic etching of the dummy gate lines.

• Step 2.6.3 (RTA): Source/drain dopant activation anneal conducted with the dummy gate in place (Figure 8).

• Subsequent Steps: Dummy poly pull etch followed by dual work-function replacement metal gate ALD deposition.

(b) Process Flow Classification:

The 14 nm FinFET process is a Gate-Last (Replacement Metal Gate / RMG) architecture. The dummy poly gate acts as a place-holder during high-temperature source/drain epitaxy (SiGe for PMOS, SiC for NMOS) and RTA, protecting channel integrity.

(c) Gate Metal Materials for PMOS and NMOS:

Advanced FinFET CMOS requires dual work-function metal gate stacks to set low and symmetric threshold voltages (V_{th}) without excessive channel doping:

• NMOS Gate Metal: TiAl (Titanium Aluminum) or TiAlC — A low work-function metal (~4.1 to 4.3 eV) energetically aligned near the silicon conduction band (E_c).

• PMOS Gate Metal: TiN (Titanium Nitride) or TaN — A high work-function metal (~4.9 to 5.1 eV) energetically aligned near the silicon valence band (E_v).

5. 14 NM FINFET SACRIFICIAL GATE & POST-RTA STRUCTURE IMAGES

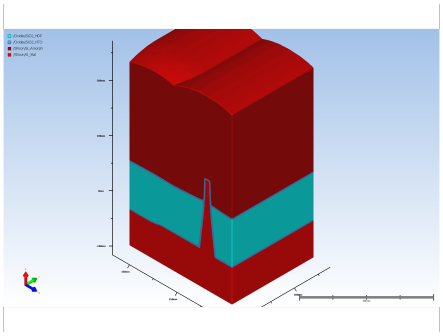


Figure 7 : Sacrificial Gate Deposition in 14 nm FinFET (Step 2.5.3)

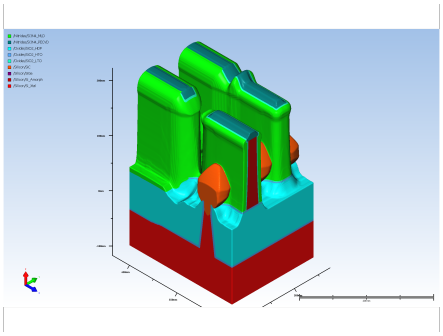


Figure 8 : 14 nm FinFET Structure Post-RTA Prior to RMG (Step 2.6.3)